

OMASANJUWA ANDREW ETSANO

Software Engineer | AI Evaluation | Technical Documentation

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SUMMARY

Software engineer and AI evaluation contractor with hands-on experience in full-stack development, AI coding-agent benchmark design, enterprise system QA/UAT, and technical documentation. Shipped a multi-tenant SaaS platform solo, authored evaluation tasks for autonomous coding agents on production Rust codebases, and supported national public health systems in Nigeria through development, testing, documentation, and field coordination. Currently completing an MS in Data Science and AI.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, SQL, HTML/CSS, Rust (evaluation task authoring, agent-assisted)
- **Frontend:** React 18/19, Next.js 16, TanStack Start, Vite, Tailwind CSS, shadcn/ui, Radix UI, TanStack Query, Framer Motion, GSAP, Recharts, Leaflet, Chart.js
- **Backend & Data:** Node.js, Django/DRF, Supabase (PostgreSQL, Auth, Edge Functions, Row Level Security), SQLite, REST APIs, DHIS2, KoboToolbox
- **AI & Evaluation:** Agent benchmark design, test-suite and adversarial probe design, difficulty calibration, RAG pipelines, multi-provider LLM integration (OpenAI, Gemini), statistical testing (Wilcoxon, Cohen's kappa, Gwet's AC1), AI-assisted development (Claude Code, Cursor, Antigravity)
- **DevOps & Tooling:** Git/GitHub, Docker, CI sandbox debugging, Linux, Bash, PowerShell, Playwright, Vercel
- **Documentation:** Role-based workflow guides, SRS structuring, Mermaid flowcharts, Moodle (Aiken/GIFT), Word/DOCX scripting

EXPERIENCE

AI Coding-Agent Evaluation, Task Author (Contract) | *Shipd, Project Olympus*

2026

- Design and build benchmark tasks that measure autonomous AI coding agents against real, unfinished work in large open-source codebases.
- Authored end-to-end tasks on production Rust projects: a reference implementation of an unimplemented feature, a large test suite, a containerised offline test harness, and a specification precise enough to solve from without giving the tests away.
- Implemented database fixture capture and snapshot diffing for SQLite and PostgreSQL, covering catalogue introspection, round-tripping of 30+ column types, foreign-key-ordered replay, cyclic table handling, and primary-key-matched row diffs.
- Calibrated difficulty empirically: ran 10 agents per revision, replayed saved patches against candidate test changes, and predicted pass rates before committing, holding them inside the required 10 to 50% band across many rounds.
- Built a 30+ case adversarial probe suite that attacks passing agent solutions the way the review panel does, surfacing specification gaps before external review.
- Cleared a multi-stage review funnel (originality screening, fairness and specification audits, AI reviewer, false-positive panel). One task approved at human review; a second carried through the same funnel.
- Diagnosed failures inside a sealed CI sandbox (toolchain resolution, offline builds, per-run PostgreSQL provisioning, JUnit reporting) and verified every fix against real agent runs rather than taking it on trust.

DevOps Intern | *GeorgeTown Global Health Nigeria, Abuja*

Jan 2026 – Present

- **Knight Data Repository:** Built the frontend and REST API integrations for a centralized health-dataset platform (vanilla JavaScript, Tailwind CSS, Leaflet, Chart.js), with an interactive geographic and categorical dashboard, secure authentication, activity tracking, and archival workflows. Reviewed its Technical Report, processing tracked changes and comments.
- **NPHCDA ERP (QA/UAT and documentation):** Wrote workflow documentation for procurement, HR, finance, and other modules cross-referenced to the master SRS, including a procurement lifecycle guide with role-based colour coding. Reformatted the full SRS into a clean Word document with Node.js docx scripting, built interactive testing dashboards covering 70+ requirements across 13 modules, and interpreted Mermaid flowcharts for UAT. Tracked VLE restoration progress and wrote a Python FFmpeg batch compression script for training videos.
- **PHC training content:** Converted quiz content from bilingual Word documents into Moodle-importable Aiken and GIFT formats across five languages (English, Hausa, Igbo, Yoruba, Pidgin), covering modules such as nutrition, family planning, and mental health plus 20 pre/post-test question banks. Cross-checked every stated answer against its explanation and the local-language key, and flagged mismatches.
- **DHIS2 and field data:** Mapped three national DR-TB paper forms (referral/transfer, identity hand card, in-patient discharge) to DHIS2 tracker program metadata and program rules. On the DNO nationwide health facility assessment (HIV, TB, and hepatitis diagnostic services), worked in the Command and Control Unit for Edo, Bayelsa, and Cross River: coordinated and supervised field assessors, validated KoboToolbox submissions for duplicates, GPS accuracy, and invalid records, and wrote the state assessment report.
- **Research and communication:** Conducted and transcribed Key Informant Interviews for the ScanForm post-pilot assessment, and built a presentation on a healthcare waste management baseline assessment across 42 facilities in Northern Nigeria.

- **Also:** Ran UAT and compiled bug logs for an enterprise data portal with the digital health innovations team, supported Linux systems, and led testing and publication of Android apps on the Google Play Store (App Bundle setup, beta testing, store assets).

Frontend Developer Intern & Content Reviewer | Flexisaf Edusoft, Abuja

Jan 2024 – Jul 2024

- Joined an existing production codebase and built UI components with HTML, CSS, JavaScript, and React, collaborating through Git in Agile workflows.
- Reviewed and quality-checked AI-generated quiz content, improving accuracy across 200+ items.

PROJECTS

BookDex | Multi-tenant SaaS booking and invoicing platform, bookdex.site

2025 – Present

- Designed and shipped solo a booking, client management, and invoicing platform for African service-based businesses: React 18, TypeScript, Vite, TanStack Query, shadcn/ui, Tailwind CSS, Supabase (PostgreSQL, Auth).
- Enforced tenant isolation and staff permissions with PostgreSQL Row Level Security, and built Edge Functions (TypeScript on Deno) for staff invites, booking notifications, and email and WhatsApp confirmations, plus client-side PDF invoice generation.
- Hardened Supabase security, audited the codebase, built the animated production landing page, worked on Nigeria-focused SEO, and deployed on Vercel with Core Web Vitals tracking.

ZOLIG Consulting website | Next.js corporate site, live on Vercel

2026

- Built a corporate website for a Nigerian construction, procurement and protective-equipment company in Next.js 16, React 19 and TypeScript, with service, project, gallery and journal pages and an enquiry form. Prepared it for native Vercel deployment by replacing the Cloudflare-specific pieces of the original export. Built with AI-assisted tooling.

Bomas Academy website | School site and admin redesign, TanStack Start and Supabase

2026

- Redesigned the public site and staff admin area of an existing school codebase (React 19, TanStack Start, Supabase, Tailwind v4, Framer Motion): navigation and mobile menu, admissions and news pages, and admin screens for content, staff, news and gallery. Updated the build to a TanStack Start version Vercel accepts.

ShefGuide | AI academic-support platform, collaborative MSc project (in progress)

- Built collaboratively. Personally built the React and TypeScript frontend, the dual-source RAG pipeline (vector embeddings, cosine similarity, provider-agnostic GPT-4o-mini and Gemini switching), and the evaluation harness with statistical testing and an 8-attack prompt-injection test suite. Backend and auth were built by a teammate.

sqlglot (open source) | SQL optimizer contribution

- Implemented a simplify_joins optimizer rule following the codebase's conventions, validating correctness against DuckDB.

EDUCATION

M.S., Data Science and AI, Campbellsville University

Jan 2026 – Present

- Coursework in AI ethics, AI applications, and AI in supply chain. Wrote a Big Data Capstone systematic literature review and a five-chapter APA research report on AI ethics in decision-making.

B.Sc., Software Engineering, Babcock University

Oct 2021 – Jul 2025

CERTIFICATIONS

- IBM DevOps & Software Engineering Professional Certificate (Coursera)
- Python for Data Science, AI & Development (Coursera)
- Containers with Docker, Kubernetes & OpenShift (Coursera)